

Joon-Hwi Kim (김준휘)

Ph.D. in Physics · Walter Burke Institute for Theoretical Physics, Pasadena, CA 91125, USA

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Google Scholar: <https://scholar.google.com/citations?user=A15RZN4AAAAI>

InspireHEP: <https://inspirehep.net/authors/1926101> | Github: <https://github.com/Joonhwi>

Research Interests & Profile

A theoretical physicist with expertise in Quantum Field Theory, Differential Geometry, Graph Theory, and Generalized Symmetries, with interests in applications to AI and ML.

Education

Ph.D. in Physics, Caltech, 4.20/4.3 (Pasadena, CA, USA)	09/2021-06/2026
Advisor: Clifford Cheung	
Thesis: <i>Reimagining Gravity: Generalized Symmetries, Double Copy, and Spinning Black Holes</i>	
B.S. in Physics, Seoul National University, 4.03/4.3 (Seoul, Korea)	02/2017-02/2021
Advisor: Sangmin Lee	
Seoul Science High School for the Gifted, 4.20/4.3 (Seoul, Korea)	02/2014-02/2017

Honors and Awards

Gold Medal in the 46th International Physics Olympiad (Tata Institute of Fundamental Research)	07/2015
Gold Prize in the 5th Hanwha Science Challenge (₩20,000,000, Hanwha Co., Ltd.)	08/2015
Korea Talent Award (Korea Foundation for Advancement of Science & Technology, 50 individuals/nation)	11/2016
Presidential Science Scholarship, Korea Student Aid Foundation (Full tuition)	03/2017-02/2021
Research Grant, Ilju Academy of Science of Culture (\$120,000)	08/2021-07/2025
Nominee for Christy Thesis Prize (Best Theoretical Physics Thesis) (Caltech)	06/2026

Publications

[2405.17056] Massive twistor worldline in electromagnetic fields	30
JHK, J.-W. Kim, S. Lee <i>J. High Energy Phys.</i> 8 (2024) 080	
[2410.22988] Classical eikonal from Magnus expansion	26
JHK, J.-W. Kim, S. Kim, S. Lee	
<i>J. High Energy Phys.</i> 1 (2025) 111 (Mathematica codes available)	
[2403.01837] Generalized symmetry in dynamical gravity	23
C. Cheung, M. Derda, JHK, V. Nevoa, I. Rothstein, N. Shah	
<i>J. High Energy Phys.</i> 10 (2024) 007	
[2309.11886] Asymptotic spinspace-time	16
JHK <i>Physical Review D</i> 111 (2025) 10, 105011	
[2412.19611] Newman-Janis algorithm from Taub-NUT instantons	15
JHK <i>Physical Review Letters</i> 136 (2026) 23, 231401	

[2405.09518]	Single Kerr-Schild metric for Taub-NUT instanton JHK <i>Physical Review D</i> 111 (2025) 2, L021703	20
[2301.06203]	Symplectic perturbation theory in massive twistor space: a zig-zag theory of massive spinning particles JHK, S. Lee	18
[2102.07063]	The relativistic spherical top as a massive twistor JHK, J.-W. Kim, S. Lee <i>J. Phys. A</i> 54 (2021) 33, 335203	14
[2511.07387]	Manifest symplecticity in classical scattering JHK <i>J. High Energy Phys.</i> 5 (2026) 287	9
[2512.23697]	Note on the Kerr spinning-particle equations of motion JHK	7
[2509.23600]	Geodesic deviation to all orders via a tangent bundle formalism JHK <i>General Relativity and Gravitation</i> 58 (2026) 7 75 (Mathematica codes available)	6
[2509.06058]	Worldline formalism in phase space JHK	5
[2602.19495]	The Kerr two-twistor particle JHK	4
[2603.07537]	The Kerr-Newman two-twistor particle JHK	3
[2512.23100]	Phase space formulation of S-matrix JHK	3
[2603.27353]	Universality in relativistic spinning particle models JHK, S. Lee (Under review, <i>J. High Energy Phys.</i>)	2
[2510.23080]	Double copy and the double Poisson bracket JHK	2
[2603.21934]	Covariant symplectic geometry of classical particles JHK	1
[2605.25473]	The diagrammar of quantum Magnusian L. Guo, JHK, J.-W. Kim, S. Kim, S. Lee, J.-R. Li (Mathematica codes available)	0
[2512.21210]	Twisted Feynman integrals: from generating functions to spin-resummed post-Minkowskian dynamics JHK, J.-W. Kim, J. Lim (Under review, <i>J. High Energy Phys.</i>)	0

Invited Talks

Twistor Particle Programme Rebooted: A Zig-Zag Theory of Massive Spinning Particles QFT-Relativity Seminars Oxford University (Oxford, UK)	11/17/2023
One-Form Symmetry in Dynamical Gravity Theoretical Elementary Particle (TEP) Physics Seminar UCLA (Los Angeles, California, USA)	04/18/2024
Newman-Janis Shift and Beyond Séminaire Amplitudes et Gravitation sur l'Yvette IHÉS (Bures-sur-Yvette, France)	01/15/2025
Building Black Holes from Taub-NUT Instantons Self-Force Group Seminar Series University of Southampton (Hampshire, UK)	02/05/2025
Worldline Formalism in Phase Space Worldline Seminars	10/28/2025

Conference Talks

Massive Twistor Theory for Spinning Black Holes Simons Collaboration on Celestial Holography Annual Meeting, Poster Session Simons Foundation (New York, USA)	04/16/2026
Double Copy from Ambitwistor Worldline	11/08/2025

California Amplitudes November Meeting 2025, II UCLA (Los Angeles, California, USA)

Spinning Black Holes as Molecules of Chiral Dyons 05/31/2025

California Amplitudes November Meeting 2025, I UC Davis (Los Angeles, California, USA)

Newman-Janis Algorithm from Taub-NUT Instantons 04/05/2025

41st Pacific Coast Gravity Meeting CSUF (Fullerton, California, USA)

Spinning Black Hole Amplitudes from Complexified Equivalence Principle 06/12/2024

Amplitudes 2024, Gong Show IAS (Princeton, New Jersey, USA)

Spinspacetime: Unification of Spacetime and Spin into a Complex Geometry 03/31/2023

39th Pacific Coast Gravity Meeting Caltech (Pasadena, California, USA)

Symplectic Perturbation Theory 03/12/2023

California Amplitudes Meeting 2023 UCSD (San Diego, California, USA)

Selected Academic Travels and Activities

Mathematical Structures in Scattering Amplitudes 08/2024

Galileo Galilei Institute for Theoretical Physics (Florence, Italy) (Invitation by Lionel Mason)

Symmetries and Anomalies: a Modern Take 06-07/2024

Institut des Hautes Études Scientifiques (Bures-sur-Yvette, France)

Amplitudes 2024 Institute of Advanced Study (Princeton, New Jersey, USA) 06/2024

Summer School on Numerical Relativity: Black Hole Imaging with Supercomputer Assistance 06/2019

APCTP-NIMS-KISTI-UNIST-KASI (Seoul, Korea)

Coursework

Caltech

- Introduction to String Theory, Ph 250ab Winter, Spring 24-25
- Introduction to Knot Theory and Quantum Topology, Ma 146b Winter 23-24
- Topics in Mathematical Physics, Ma 148a Fall 22-23
- Advanced Mathematical Methods of Physics, Ph 229b; Elementary Particle Theory, Ph 230b Spring 21-22
- Riemannian Geometry, Ma 157b; Relativity, Ph 236b Winter 21-22
- Mathematical Methods of Physics, Ph 129a; Advanced Mathematical Methods of Physics, Ph 229a Fall 21-22
- Relativistic Quantum Field Theory, Ph 205a; Elementary Particle Theory, Ph 230a Fall 21-22
- Introduction to Topological Field Theory, Ph 232 Fall 21-22

Seoul National University

- Particle Physics (Graduate); Classical Mechanics (Graduate); Quantum Mechanics I (Graduate) 2020
- Properties of Solids 2020
- Quantum Fields and Strings on Curved Spacetime (Graduate, KIAS SITP) 2019
- Quantum Field Theory I, II (Graduate); Elementary Nuclear and Particle Physics (Graduate) 2019
- Theory of Lie Groups (Graduate); Statistical Mechanics and Thermal Physics 2019
- Cognitive Linguistics; Music Perception and Cognition; Perception of Visual Art 2017-2018
- Computational Linguistics and Natural Language Processing; Artificial Intelligence for Physicists 2018
- Fluid Dynamics; Relativity and Spacetime 2017

Teaching & Service

Teaching Assistant

- Caltech (2022-2026)
 - Theoretical Cosmology and Astroparticle Physics, Ph 235b Spring 25-26

- Mathematical Methods of Physics, Ph 129b Winter 25-26
- General Relativity, Ph 236a Fall 25-26
- Topics in Classical Physics (electrodynamics and relativity), Ph 106c Spring 24-25
- Topics in Classical Physics (electromagnetism), Ph 106b Winter 24-25
- Topics in Classical Physics (mechanics), Ph 106a Fall 24-25
- Quantum Mechanics, Ph 125c Spring 23-24
- General Relativity, Ph 236b, prof. Saul Teukolsky Winter 23-24
- General Relativity, Ph 236a Fall 23-24
- Candidacy Exam Fitness, Ph 201 Spring 22-23
- Quantum Mechanics, Ph 2b Winter 22-23
- Classical Mechanics, Ph 1a Fall 22-23
- ❑ Korean Physics Society (2017-2019)
 - Theoretical and Experimental Problem Solving for International Physics Olympiad

Volunteering

- ❑ Caltech (2022-2025)
 - Future Ignited Conference 09/2022, 2023
 - Caltech Campus Tour 11/2022, 2023, 2024, 2025

Lecturer

- ❑ Physics and the Imagination 03-12/2017
 - 10 lectures on mathematical physics (quaternions, spinors and Clifford algebra, matrix representations, differential forms, de Rham cohomology); Refs. Schutz (1980), Nakahara (2003), Doran (2003)
- ❑ Escher's Studio 03-12/2017
 - Interdisciplinary program that explored the intersections between **mathematics** (group theory, homology, complex analysis), **physics** (symmetry, representation theory, gauge theory), **computer science** (Javascript programming, generative art), **visual art** (tessellations, planar compositions, illusions, impossible figures, perspectives, drawing on nonlinear grids), and **music** (musical tessellations in chord space, Shepard tone).

Referee Role

- ❑ European Physical Journal C (2025), International Journal of Geometric Methods in Modern Physics (2026), International Journal of Theoretical Physics (2026)

Media

How to turn the complex mathematics of vector calculus into simple pictures. [MIT Technology Review](#), Nov 2019.

Skills

<i>Programming</i>	Mathematica, Symbolic Tensor Manipulation via xAct, Python, JavaScript, HTML
<i>Miscellaneous</i>	Adobe Illustrator, Adobe Photoshop, Rhinoceros 3D, MIDI Sequencers, LaTeX
<i>Languages</i>	Korean (Native), English (Fluent)

References

Clifford Cheung (ccheung@caltech.edu)

Professor, Walter Burke Institute for Theoretical Physics, Caltech

Director, Leinweber Forum for Theoretical Physics

Sangmin Lee (sangminlee@kias.re.kr)

Professor, Korea Institute for Advanced Study

Julio Parra-Martinez (julio@ihes.fr)

Professor, Institut des Hautes Études Scientifiques (French Institute for Advanced Study)